

Exponentials & Logarithms

ANSWER KEY



Laws of logarithms

- 1 Simplify $\log_2 8 + \log_2 4$ to a single number.
 $\log_2 8 + \log_2 4 = \log_2 (8 \times 4) = \log_2 32 = 5.$
- 2 Express $2\ln x - \ln(x - 1)$ as a single logarithm.
 $2\ln x - \ln(x - 1) = \ln(x^2) - \ln(x - 1) = \ln\left(\frac{x^2}{x - 1}\right).$
- 3 Simplify $\log_3 54 - \log_3 2$ to a single number.
 $\log_3 54 - \log_3 2 = \log_3\left(\frac{54}{2}\right) = \log_3 27 = 3.$

Solving exponential equations using logarithms

- 4 Solve $5^x = 20$, giving your answer to three decimal places.
Take $\ln$ of both sides: $x\ln 5 = \ln 20$. $x = \frac{\ln 20}{\ln 5} = \frac{2.995732}{1.609438} \approx 1.861.$
- 5 Solve $3^{2x-1} = 10$, to three decimal places.
 $(2x - 1)\ln 3 = \ln 10 \Rightarrow 2x - 1 = \frac{\ln 10}{\ln 3} \approx 2.095903$. So $2x \approx 3.095903$, giving $x \approx 1.548.$
- 6 Solve $7^{3x} = 50$, to three decimal places.
 $3x\ln 7 = \ln 50 \Rightarrow x = \frac{\ln 50}{3\ln 7} = \frac{3.912023}{5.837731} \approx 0.670.$

Solving logarithmic equations

- 7 Solve $\ln(2x - 1) = 3$, to three decimal places.
 $2x - 1 = e^3 \approx 20.0855$. So $2x \approx 21.0855$, giving $x \approx 10.543.$
- 8 Solve $\log_2 (x + 3) + \log_2 (x - 1) = 5$, rejecting any value outside the domain.
 $\log_2 [(x + 3)(x - 1)] = 5 \Rightarrow (x + 3)(x - 1) = 32 \Rightarrow x^2 + 2x - 3 = 32 \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x + 7)(x - 5) = 0.$
Domain requires $x > 1$, so $x = -7$ is rejected. Solution: $x = 5.$
- 9 Solve $\ln x + \ln(x + 3) = \ln 10$, rejecting any value outside the domain.
 $\ln [x(x + 3)] = \ln 10 \Rightarrow x(x + 3) = 10 \Rightarrow x^2 + 3x - 10 = 0 \Rightarrow (x + 5)(x - 2) = 0.$ Domain requires $x > 0$, so $x = -5$ is rejected. Solution: $x = 2.$

Properties of the exponential and natural log functions

- 10 Simplify $e^{\ln 7}$ and $\ln(e^4)$, using the fact that e^x and $\ln x$ are inverse functions.
 $e^{\ln 7} = 7$ and $\ln(e^4) = 4$, since e^x and $\ln x$ undo each other.
- 11 Solve $e^{2x} = 15$, to three decimal places.
 $2x = \ln 15 \approx 2.708050$, so $x \approx 1.354$.

Graphs of exponential and logarithmic functions

- 12 The graph shows $y = e^x$ and $y = \ln x$. State how the two curves are related geometrically, and state the equation of the asymptote of each.
 The two curves are reflections of each other in the line $y = x$, since $y = \ln x$ is the inverse function of $y = e^x$. The asymptote of $y = e^x$ is $y = 0$ (the x -axis); the asymptote of $y = \ln x$ is $x = 0$ (the y -axis).
- 13 The graph shows $y = \ln(x - 2)$. State its domain and the equation of its vertical asymptote.
 The domain is $x > 2$ (the argument $x - 2$ must be positive). The vertical asymptote is $x = 2$.

Exponential growth and decay models

- 14 A bacteria culture grows according to $N(t) = 200e^{0.15t}$, where t is in hours. Find the population after 10 hours, to the nearest whole number.
 $N(10) = 200e^{1.5} = 200(4.481689) \approx 896$.
- 15 Using the same model $N(t) = 200e^{0.15t}$, find how long it takes the population to double, to three decimal places.
 $200e^{0.15t} = 400 \Rightarrow e^{0.15t} = 2 \Rightarrow 0.15t = \ln 2 \approx 0.693147 \Rightarrow t \approx 4.621$ hours.

Simultaneous equations involving exponentials and logarithms

- 16 Solve simultaneously $\log_2 x + \log_2 y = 5$ and $x - y = 4$, for $x, y > 0$.
 From the first equation, $xy = 2^5 = 32$. Substituting $x = y + 4$:
 $(y + 4)y = 32 \Rightarrow y^2 + 4y - 32 = 0 \Rightarrow (y + 8)(y - 4) = 0$. Since $y > 0$, $y = 4$, giving $x = 8$.
- 17 Solve simultaneously $\ln x - \ln y = 1$ and $x + y = 11$, giving x and y to three decimal places.
 From the first equation, $\ln\left(\frac{x}{y}\right) = 1 \Rightarrow \frac{x}{y} = e$. Substituting:
 $ey + y = 11 \Rightarrow y(e + 1) = 11 \Rightarrow y = \frac{11}{e + 1} \approx \frac{11}{3.718282} \approx 2.958$. Then $x = 11 - y \approx 8.042$.

The change of base formula

- 18 Evaluate $\log_5 30$ using the change of base formula, to three decimal places.
 $\log_5 30 = \frac{\ln 30}{\ln 5} = \frac{3.401197}{1.609438} \approx 2.113$.

- 19 Solve $2^x = 3^{x-1}$ using logarithms, to three decimal places.

Take $\ln$ of both sides:

$$x \ln 2 = (x - 1) \ln 3 \Rightarrow x \ln 2 - x \ln 3 = -\ln 3 \Rightarrow x(\ln 2 - \ln 3) = -\ln 3 \Rightarrow x = \frac{\ln 3}{\ln 3 - \ln 2} = \frac{1.098612}{0.405465} \approx 2.710$$

- 20 Evaluate $\log_7 3 \times \log_3 7$ without a calculator, explaining why the product is exactly 1.

By the change of base formula, $\log_7 3 = \frac{\ln 3}{\ln 7}$ and $\log_3 7 = \frac{\ln 7}{\ln 3}$. Their product is $\frac{\ln 3}{\ln 7} \times \frac{\ln 7}{\ln 3} = 1$, since the $\ln 3$ and $\ln 7$ factors cancel completely — this holds for any pair of reciprocal-base logarithms $\log_a b \times \log_b a$.

A well-designed word problem: radioactive decay

- 21 This question has three parts. A radioactive isotope decays according to $A(t) = A_0 e^{-kt}$, where A_0 is the initial mass (grams) and t is time in years. (a) Given that the isotope has a half-life of 12 years, find the decay constant k , to four decimal places. (b) A sample starts with $A_0 = 80$ grams. Find the mass remaining after 30 years, to two decimal places. (c) Find how long it takes the sample to decay to 5 grams.

(a) $\frac{A_0}{2} = A_0 e^{-12k} \Rightarrow e^{-12k} = 0.5 \Rightarrow -12k = \ln 0.5 \approx -0.693147 \Rightarrow k \approx 0.0578$. (b)

$A(30) = 80e^{-0.0578(30)} = 80e^{-1.7329} \approx 80(0.1760) \approx 14.1$ grams. (c) Solve

$80e^{-0.0578t} = 5 \Rightarrow e^{-0.0578t} = \frac{5}{80} = \frac{1}{16}$. Since $\frac{1}{16} = \left(\frac{1}{2}\right)^4$, this corresponds to exactly 4 half-lives, so $t = 4 \times 12 = 48$ years.